CS251 - Homework 2

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Lemma 1. $t_1 = t_2 \vdash t_2 = t_1$.

1.	$t_1 = t_2$	premise
2.	$t_1 = t_1$	=i
3.	$t_2 = t_1$	=e 1, 2

Problem 2.3.1. Prove the validity of the following sequents using, among others, the rule =i and =e. Make sure that you indicate for each application of =e what the rule instances ϕ, t_1 and t_2 are.

a. $(y = 0) \wedge (y = x) \vdash 0 = x$.

b. $t_1 = t_2 \vdash (t + t_2) = (t + t_1)$.

c. $(x = 0) \vee ((x + x) > 0) \vdash (y = (x + x)) \rightarrow ((y > 0) \vee (y = (0 + x)))$.

a.

1.	$(y = 0) \wedge (y = x)$	premise
2.	$y = 0$	$\wedge e_1$ 1
3.	$y = x$	$\wedge e_2$ 1
4.	$0 = x$	=e 2, 3

The rule of inference =e at line 4 is instantiated with ϕ being $t_1 = x$, t_1 being y, and t_2 being 0.

b. Due to notation clash, we redefine the equality elimination instantiation parameters t_1 and t_2 to t_1' and t_2', respectively.

1.	$t_1 = t_2$	premise
2.	$t + t_1 = t + t_1$	=i
3.	$t + t_2 = t + t_1$	=e 1, 2

The rule of inference =e at line 3 is instantiated with ϕ being $t + t_1' = t + t_1$, t_1' being t_1, and t_2' being t_2.

c.

1.	$(x = 0) \vee ((x + x) > 0)$	premise
2.	$y = (x + x)$	assumption
3.	$x = 0$	assumption
4.	$y = (0 + x)$	=e 3, 2
5.	$(y > 0) \vee (y = (0 + x))$	$\vee$i$_4$ 4
6.	$(x + x) > 0$	assumption
7.	$(x + x) = y$	(Lemma 1), 2
8.	$y > 0$	=e 7, 6
9.	$(y > 0) \vee (y = (0 + x))$	$\vee$i$_1$ 8
10.	$(y > 0) \vee (y = (0 + x))$	$\vee$e 1, 3–5, 6–9
11.	$(y = (x + x)) \to ((y > 0) \vee (y = (0 + x)))$	$\to$i 2–10

The rule of inference =e at line 4 is instantiated with ϕ being $y = (t_1 + x)$, t_1 being x, and t_2 being 0.

The rule of inference =e at line 3 in the Lemma 1 instantiation at line 7 is instantiated with ϕ being $t_1 = y$, t_1 being y, t_2 being $x + x$.

The rule of inference =e at line 8 is instantiated with ϕ being $t_1 > 0$, t_1 being $x + x$, and t_2 being y.

Problem 2.3.4.

a. Find a (propositional) proof for $\phi \to (q_1 \wedge q_2) \vdash (\phi \to q_1) \wedge (\phi \to q_2)$.

b. Find a (predicate) proof for $\phi \to \forall x Q(x) \vdash \forall x(\phi \to Q(x))$, provided that x is not free in ϕ.

c. Find a proof for $\forall x(P(x) \to Q(x)) \vdash \forall x P(x) \to \forall x Q(x)$.

a.

1.	$\phi \to (q_1 \wedge q_2)$	premise
2.	ϕ	assumption
3.	$q_1 \wedge q_2$	$\to$e 1, 2
4.	q_1	$\wedge$e$_1$ 3
5.	$\phi \to q_1$	$\to$i 2–4
6.	ϕ	assumption
7.	$q_1 \wedge q_2$	$\to$e 1, 6
8.	q_2	$\wedge$e$_2$ 7
9.	$\phi \to q_2$	$\to$i 6–8
10.	$(\phi \to q_1) \wedge (\phi \to q_2)$	$\wedge$i 5, 9

b.

1.	$\phi \to \forall x Q(x)$	premise
2.	x_0	assumption
3.	ϕ	assumption
4.	$\forall x Q(x)$	$\to$e 1, 2
5.	$Q(x_0)$	$\forall$e 4
6.	$\phi \to Q(x_0)$	$\to$i 3–5
7.	$\forall x(\phi \to Q(x))$	$\forall$i 2–6

c.

1.	$\forall x(P(x) \to Q(x))$	premise
2.	$\forall x P(x)$	assumption
3.	x_0	assumption
4.	$P(x_0) \to Q(x_0)$	$\forall$e 1
5.	$P(x_0)$	$\forall x$e 2
6.	$Q(x_0)$	$\to$e 4, 5
7.	$\forall x Q(x)$	$\forall$i 3–6
8.	$\forall x P(x) \to \forall x Q(x)$	$\to$i 2–7

Problem 2.3.6. Provide proofs for the following sequents:

a. $\forall x P(x) \vdash \forall y P(y)$; using $\forall x P(x)$ as a premise, your proof needs to end with an application of $\forall$i which requires the formula $P(y_0)$.

b. $\forall x(P(x) \to Q(x)) \vdash (\forall x \neg Q(x)) \to (\forall x \neg P(x))$.

c. $\forall x(P(x) \to \neg Q(x)) \vdash \neg(\exists x(P(x) \wedge Q(x)))$.

a.

1.	$\forall x P(x)$	premise
2.	y_0	assumption
3.	$P(y_0)$	$\forall$e 1
4.	$\forall y P(y)$	$\forall$i 2–3

b.

1.	$\forall x(P(x) \to Q(x))$	premise
2.	$\forall x \neg Q(x)$	assumption
3.	x_0	assumption
4.	$\neg Q(x_0)$	$\forall$e 2
5.	$P(x_0) \to Q(x_0)$	$\forall$e 1
6.	$\neg P(x_0)$	MT 5, 4
7.	$\forall x \neg P(x)$	
8.	$(\forall x \neg Q(x)) \to (\forall x \neg P(x))$	$\to$i 2–7

c.

1.	$\forall x(P(x) \to \neg Q(x))$	premise
2.	$\neg\neg(\exists x(P(x) \land Q(x)))$	assumption
3.	$\exists x(P(x) \land Q(x))$	$\neg\neg$e 2
4.	x_0	assumption
5.	$P(x_0) \land Q(x_0)$	assumption
6.	$P(x_0)$	$\land e_1$ 5
7.	$Q(x_0)$	$\land e_2$ 5
8.	$P(x_0) \to \neg Q(x_0)$	$\forall$e 1
9.	$\neg Q(x_0)$	$\to$e 8, 6
10.	$\bot$	$\neg$i 6, 8
11.	$\bot$	$\exists$e 2, 3–9
12.	$\neg(\exists x(P(x) \land Q(x)))$	PBC 2–11

Lemma 2 (DeMorgan).

$$\neg(\phi \land \psi) \dashv\vdash \neg\phi \lor \neg\psi$$

$$\neg(\phi \lor \psi) \dashv\vdash \neg\phi \land \neg\psi$$

Proof. Well-known.

Lemma 3 ($\neg\forall$-elimination).

$$\neg\forall x\phi \dashv\vdash \exists x\neg\phi$$

Proof. Well-known.

Lemma 4 ($\neg\exists$-elimination).

$$\neg\exists x\phi \dashv\vdash \forall x\neg\phi$$

Proof. Well-known.

Problem 2.3.9. Prove the validity of the following sequents in predicate logic, where F, G, P, and Q have arity 1, and S has arity 0 (a 'propositional atom'):

b. $S \to \exists x Q(x) \vdash \exists x(S \to Q(x))$.

e. $\forall x(P(x) \lor Q(x)) \vdash \forall x P(x) \lor \exists x Q(x)$.

i. $\exists x(\neg P(x) \land \neg Q(x)) \vdash \exists x(\neg(P(x) \land Q(x)))$.

b.

1.	$S \to \exists x Q(x)$	premise
2.	x_0	assumption
3.	S	assumption
4.	$\exists x Q(x)$	$\to$e 1, 3
5.	$Q(x_0)$	assumption
6.	$S \to Q(x_0)$	$\to$i 3–5
7.	$\exists x(S \to Q(x))$	$\exists$i 6
8.	$\exists x(S \to Q(x))$	copy 7

e.

1.	$\forall x(P(x) \vee Q(x))$	premise
2.	$\neg(\forall x P(x) \vee \exists x Q(x))$	assumption
3.	$\neg\forall x P(x) \wedge \neg\exists x Q(x)$	DeMorgan 2
4.	$\neg\forall x P(x)$	$\wedge e_1$ 3
5.	$\exists x(\neg P(x))$	$\neg\exists$e 4
6.	$\neg\exists x Q(x)$	$\wedge e_2$ 3
7.	$\forall x(\neg Q(x))$	$\neg\forall$e 6
8.	x_0	assumption
9.	$\neg P(x_0)$	assumption
10.	$\neg Q(x_0)$	$\forall$e 7
11.	$\neg P(x_0) \wedge \neg Q(x_0)$	$\wedge$i 9, 10
12.	$\neg(P(x_0) \vee Q(x_0))$	DeMorgan 11
13.	$P(x_0) \vee Q(x_0)$	$\forall$e 1
14.	$\perp$	$\neg$i 13, 12
15.	$\perp$	$\exists$e 5, 8–14
16.	$\forall x P(x) \vee \exists x Q(x)$	PBC 2–15

i.

1.	$\exists x(\neg P(x) \wedge \neg Q(x))$	premise
2.	$\neg\exists x(\neg(P(x) \wedge Q(x)))$	assumption
3.	$\forall x(\neg\neg(P(x) \wedge Q(x)))$	$\neg\exists$e 2
4.	x_0	assumption
5.	$\neg P(x_0) \wedge \neg Q(x_0)$	assumption
6.	$\neg P(x_0)$	$\wedge e_1$ 5
7.	$\neg\neg(P(x_0) \wedge Q(x_0))$	$\forall$e 3
8.	$P(x_0) \wedge Q(x_0)$	$\neg\neg$e 7
9.	$P(x_0)$	$\wedge e_1$ 8
10.	$\perp$	$\neg$i 6, 9
11.	$\perp$	$\exists$e 1, 4–10
12.	$\exists x(\neg(P(x) \wedge Q(x)))$	PBC 2–11

Problem 2.3.11. The proofs of the sequents below combine the proof rules for equality and quantifiers. We write $\phi \leftrightarrow \psi$ as an abbreviation for $(\phi \to \psi) \wedge (\psi \to \phi)$. Find proofs for

b. $P(b), \forall x \forall y (P(x) \wedge P(y) \to x = y) \vdash \forall x (P(x) \leftrightarrow x = b)$.

d. $\forall x (P(x) \leftrightarrow x = b) \vdash P(b) \wedge \forall x \forall y (P(x) \wedge P(y) \to x = y)$.

b.

1.	$P(b)$	premise
2.	$\forall x \forall y (P(x) \wedge P(y) \to x = y)$	premise
3.	x_0	assumption
4.	$P(x_0)$	assumption
5.	$P(x_0) \wedge P(b)$	$\wedge$i 4, 1
6.	$\forall y (P(x_0) \wedge P(y) \to x_0 = y)$	$\forall$e 2
7.	$P(x_0) \wedge P(b) \to x_0 = b$	$\forall$e 6
8.	$x_0 = b$	$\to$e 7, 5
9.	$P(x_0) \to x_0 = b$	$\to$i 4–8
10.	$x_0 = b$	assumption
11.	$b = x_0$	(Lemma 1), 10
12.	$P(x_0)$	=e 11, 1
13.	$x_0 = b \to P(x_0)$	$\to$i 10–12
14.	$P(x_0) \leftrightarrow x_0 = b$	$\wedge$i 9, 13
15.	$\forall x (P(x) \leftrightarrow x = b)$	$\forall$i 3–14

d.

1.	$\forall x(P(x) \leftrightarrow x = b)$	premise
2.	x_0	assumption
3.	y_0	assumption
4.	$P(x_0) \wedge P(y_0)$	assumption
5.	$P(x_0)$	$\wedge e_1$ 5
6.	$P(y_0)$	$\wedge e_2$ 5
7.	$P(x_0) \leftrightarrow x_0 = b$	$\forall$e 1
8.	$P(x_0) \rightarrow x_0 = b$	$\wedge e_1$ 7
9.	$x_0 = b$	$\rightarrow$e 8, 5
10.	$P(y_0) \leftrightarrow y_0 = b$	$\forall$e 1
11.	$P(y_0) \rightarrow y_0 = b$	$\wedge e_1$ 10
12.	$y_0 = b$	$\rightarrow$e 11, 6
13.	$b = y_0$	(Lemma 1), 12
14.	$x_0 = y_0$	=e 13, 9
15.	$P(x_0) \wedge P(y_0) \rightarrow x_0 = y_0$	$\rightarrow$i 4–14
16.	$\forall y(P(x_0) \wedge P(y) \rightarrow x_0 = y)$	$\forall$i 3–15
17.	$\forall x \forall y(P(x) \wedge P(y) \rightarrow x = y)$	$\forall$i 2–16
18.	$P(b) \leftrightarrow b = b$	$\forall$e 1
19.	$P(b) \leftrightarrow \top$	
20.	$\top \rightarrow P(b)$	$\wedge e_2$ 19
21.	$P(b)$	$\rightarrow$e 20
22.	$P(b) \wedge \forall x \forall y(P(x) \wedge P(y) \rightarrow x = y)$	$\wedge$i 21, 17

Problem 2.3.13. By natural deduction, show the validity of

e. $\forall x(P(x) \vee Q(x)), \exists x \neg Q(x), \forall x(R(x) \rightarrow \neg P(x)) \vdash \exists x \neg R(x)$.

f. $\forall x(P(x) \rightarrow (Q(x) \vee R(x))), \neg \exists x(P(x) \wedge R(x)) \vdash \forall x(P(x) \rightarrow Q(x))$.

g. $\exists x \exists y(S(x, y) \vee S(y, x)) \vdash \exists x \exists y S(x, y)$.

h. $\exists x(P(x) \wedge Q(x)), \forall x(P(x) \rightarrow R(x)) \vdash \exists x(R(x) \wedge Q(x))$.

e.

1.	$\forall x(P(x) \lor Q(x))$	premise
2.	$\exists x \neg Q(x)$	premise
3.	$\forall x(R(x) \to \neg P(x))$	premise
4.	x_0	assumption
5.	$\neg Q(x_0)$	assumption
6.	$R(x_0) \to \neg P(x_0)$	$\forall$e 3
7.	$P(x_0) \lor Q(x_0)$	$\forall$e 1
8.	$P(x_0)$	assumption
9.	$\neg\neg P(x_0)$	$\neg\neg$i 8
10.	$\neg R(x_0)$	MT 6, 9
11.	$Q(x_0)$	assumption
12.	$\bot$	$\neg$i 11, 5
13.	$\neg R(x_0)$	$\bot$e 12
14.	$\neg R(x_0)$	$\lor$e 7, 8–10, 11–13
15.	$\exists x \neg R(x)$	$\exists$i 14
16.	$\exists x \neg R(x)$	$\exists$e 2, 4–15

f.

1.	$\forall x(P(x) \to (Q(x) \lor R(x)))$	premise
2.	$\neg\exists x(P(x) \land R(x))$	premise
3.	$\forall x(\neg(P(x) \land R(x)))$	$\neg\exists$e 2
4.	x_0	assumption
5.	$P(x_0)$	assumption
6.	$P(x_0) \to (Q(x_0) \lor R(x_0))$	$\forall$e 1
7.	$\neg(P(x_0) \land R(x_0))$	$\forall$e 3
8.	$Q(x_0) \lor R(x_0)$	$\to$e 6, 5
9.	$\neg P(x_0) \lor \neg R(x_0)$	DeMorgan 7
10.	$Q(x_0)$	assumption
11.	$R(x_0)$	assumption
12.	$\neg P(x_0)$	assumption
13.	$\bot$	$\neg$i 5, 12
14.	$Q(x_0)$	$\bot$e 13
15.	$\neg R(x_0)$	assumption
16.	$\bot$	$\neg$i 11, 15
17.	$Q(x_0)$	$\bot$e 16
18.	$Q(x_0)$	$\lor$e 9, 12–14, 15–17
19.	$Q(x_0)$	$\lor$e 8, 10, 11–18
20.	$P(x_0) \to Q(x_0)$	$\to$i 5–19
21.	$\forall x(P(x) \to Q(x))$	$\forall$i 4–20

g.

1.	$\exists x \exists y (S(x,y) \vee S(y,x))$	
2.	x_0	assumption
3.	$\exists y (S(x_0,y) \vee S(y,x_0))$	assumption
4.	y_0	assumption
5.	$S(x_0,y_0) \vee S(y_0,x_0)$	assumption
6.	$S(x_0,y_0)$	$\vee e_1$ 5
7.	$\exists y S(x_0,y)$	$\exists$i 6
8.	$\exists x \exists y S(x,y)$	$\exists$i 7
9.	$\exists x \exists y S(x,y)$	$\exists$e 3, 4–8
10.	$\exists x \exists y S(x,y)$	$\exists$e 1, 2–9

h.

1.	$\exists x (P(x) \wedge Q(x))$	premise
2.	$\forall x (P(x) \to R(x))$	premise
3.	x_0	assumption
4.	$P(x_0) \wedge Q(x_0)$	assumption
5.	$P(x_0)$	$\wedge e_1$ 4
6.	$Q(x_0)$	$\wedge e_2$ 4
7.	$P(x_0) \to R(x_0)$	$\forall$e 2
8.	$R(x_0)$	$\to$e 7, 5
9.	$R(x_0) \wedge Q(x_0)$	$\wedge$i 8, 6
10.	$\exists x (R(x) \wedge Q(x))$	$\exists$i 9
11.	$\exists x (R(x) \wedge Q(x))$	$\exists$e 1, 3–10

Problem 2.3.14. Translate the following argument into a sequent in predicate logic using a suitable set of predicate symbols:

> If there are any tax payers, then all politicians are tax payers. If there are any philanthropists, then all tax payers are philanthropists. So, if there are any tax-paying philanthropists, then all politicians are philanthropists.

Now come up with a proof of that sequent in predicate logic.

Define the following predicates as follows:

- $T(x)$: x is a tax payer
- $P(x)$: x is a politician
- $H(x)$: x is a philanthropist

The given argument can then be translated into the following sequent:

$$\exists x T(x) \to \forall x (P(x) \to T(x)), \exists x H(x) \to \forall x (T(x) \to H(x)) \vdash \exists x (T(x) \wedge H(x)) \to \forall x (P(x) \to H(x)).$$

This sequent is in fact true. A proof follows.

1.	$\exists x T(x) \to \forall x(P(x) \to T(x))$	premise
2.	$\exists x H(x) \to \forall x(T(x) \to H(x))$	premise
3.	$\exists x(T(x) \land H(x))$	assumption
4.	x_0	assumption
5.	$T(x_0) \land H(x_0)$	assumption
6.	$T(x_0)$	$\land e_1$ 5
7.	$\exists x T(x)$	$\exists$i 6
8.	$\forall x(P(x) \to T(x))$	$\to$e 1, 7
9.	$H(x_0)$	$\land e_2$ 5
10.	$\exists x H(x)$	$\exists$i 9
11.	$\forall x(T(x) \to H(x))$	$\to$e 2, 10
12.	$\widetilde{x}$	assumption
13.	$P(\widetilde{x}) \to T(\widetilde{x})$	$\forall$e 8
14.	$T(\widetilde{x}) \to H(\widetilde{x})$	$\forall$e 11
15.	$P(\widetilde{x})$	assumption
16.	$T(\widetilde{x})$	$\to$e 13, 15
17.	$H(\widetilde{x})$	$\to$e 14, 16
18.	$P(\widetilde{x}) \to H(\widetilde{x})$	$\to$i 15–17
19.	$\forall x(P(x) \to H(x))$	$\forall$i 12–18
20.	$\forall x(P(x) \to H(x))$	$\exists$e 3, 4–19
21.	$\exists x(T(x) \land H(x)) \to \forall x(P(x) \to H(x))$	$\to$i 3–20